REPORT 3

1. Overview:

The Smart Parking Management System was developed through a collaborative team effort, where all members contributed to the design, development, and testing of the application. The project followed a modular approach, and responsibilities were distributed to ensure efficient implementation of all components.

Each member actively participated in discussions, debugging, and integration of different modules to achieve a fully functional system.

2. Member-wise Contribution:

- **Garvit Pant (24BCE10593)**

  - Designed overall **system architecture**
  - Developed **ParkingService.java** (business logic)
  - Implemented:
    - Slot allocation logic
    - Fee calculation system

- **Drishti Agarwal (24BCE10821)**

  - Developed **FileManager.java**
  - Ensured **data persistence across sessions**
  - Implemented:
    - Serialization using ObjectInputStream & ObjectOutputStream
    - Data saving & loading

- **Aditri Sharma (24BCE10919)**

  - Designed and implemented **Java Swing UI (Main.java)**
  - Created:
    - Dashboard
    - Forms (Park Vehicle, Release Vehicle)
  - Used components:
    - JFrame, JPanel, JTable, JButton
  - Applied layout managers for clean UI

- **Yoshita Purohit (24BCE11075)**

  - Implemented **AutoSaveThread.java**
  - Ensured smooth background execution without affecting UI

- Created daemon thread for:
    - Auto-saving data every 60 seconds

◆ **Devang Srivastava (24BCE10124)**

- Implemented **ParkingException.java**
- Added error handling using try-catch
- Tested:
    - Edge cases (invalid input, full slots, etc.)
- Developed:
    - History feature
    - Report export functionality

3. Conclusion:

The project reflects strong teamwork, with all members contributing to multiple aspects of development, including design, implementation, testing, and documentation. The collaborative approach ensured smooth integration of all modules and successful completion of the project.